HotelPro

Hotel Room Booking & Administration System

Project Documentation

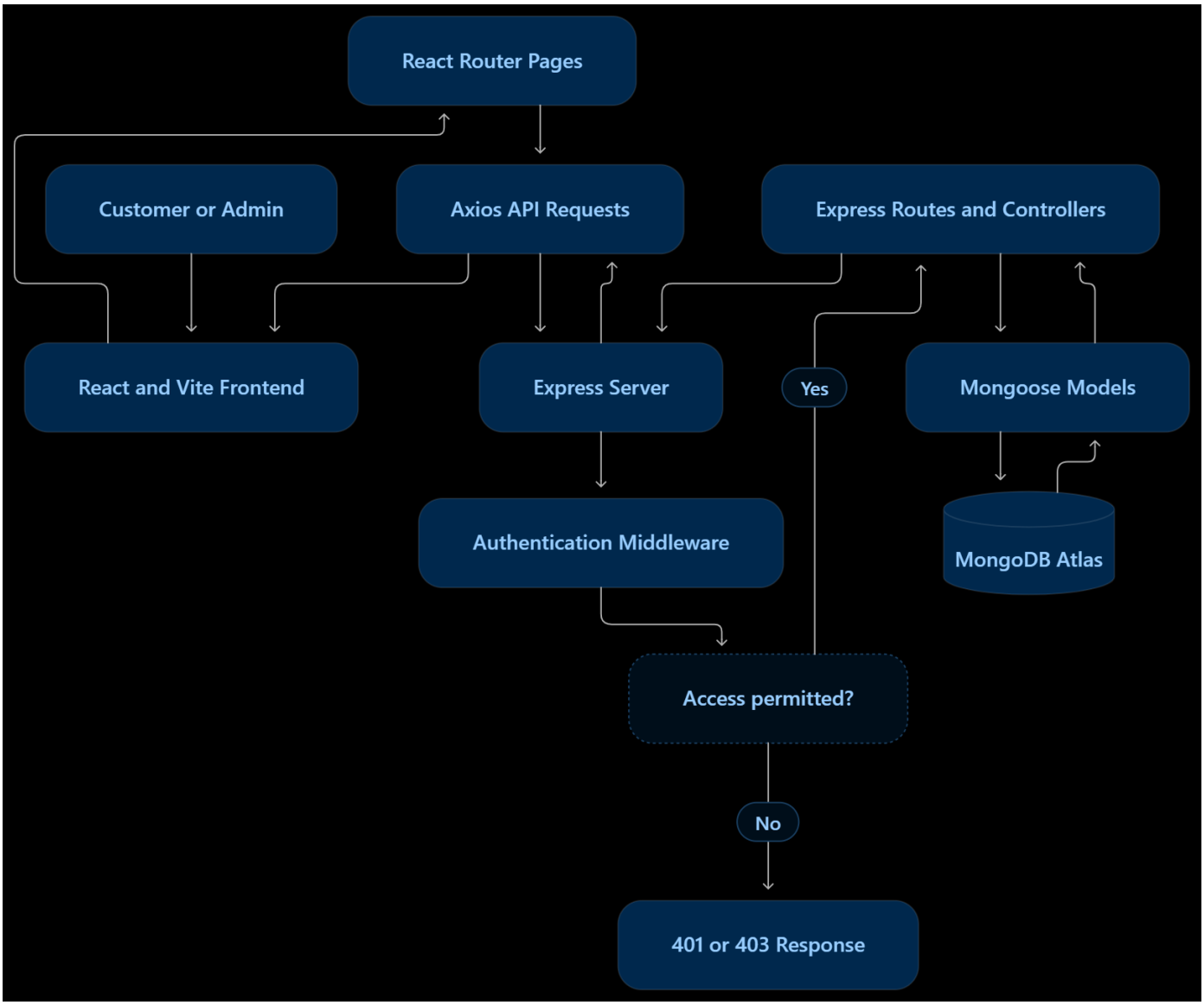
System Architecture • ER Diagram • Test-Case Report • Screenshots

Technology: React/Vite, Node.js/Express, MongoDB/Mongoose, JWT authentication, role-based authorization, bcrypt password hashing.

This document combines the architecture and data model with reported test results and screenshots supplied during project development. Screenshot evidence demonstrates the visible interface; it does not independently verify every backend test.

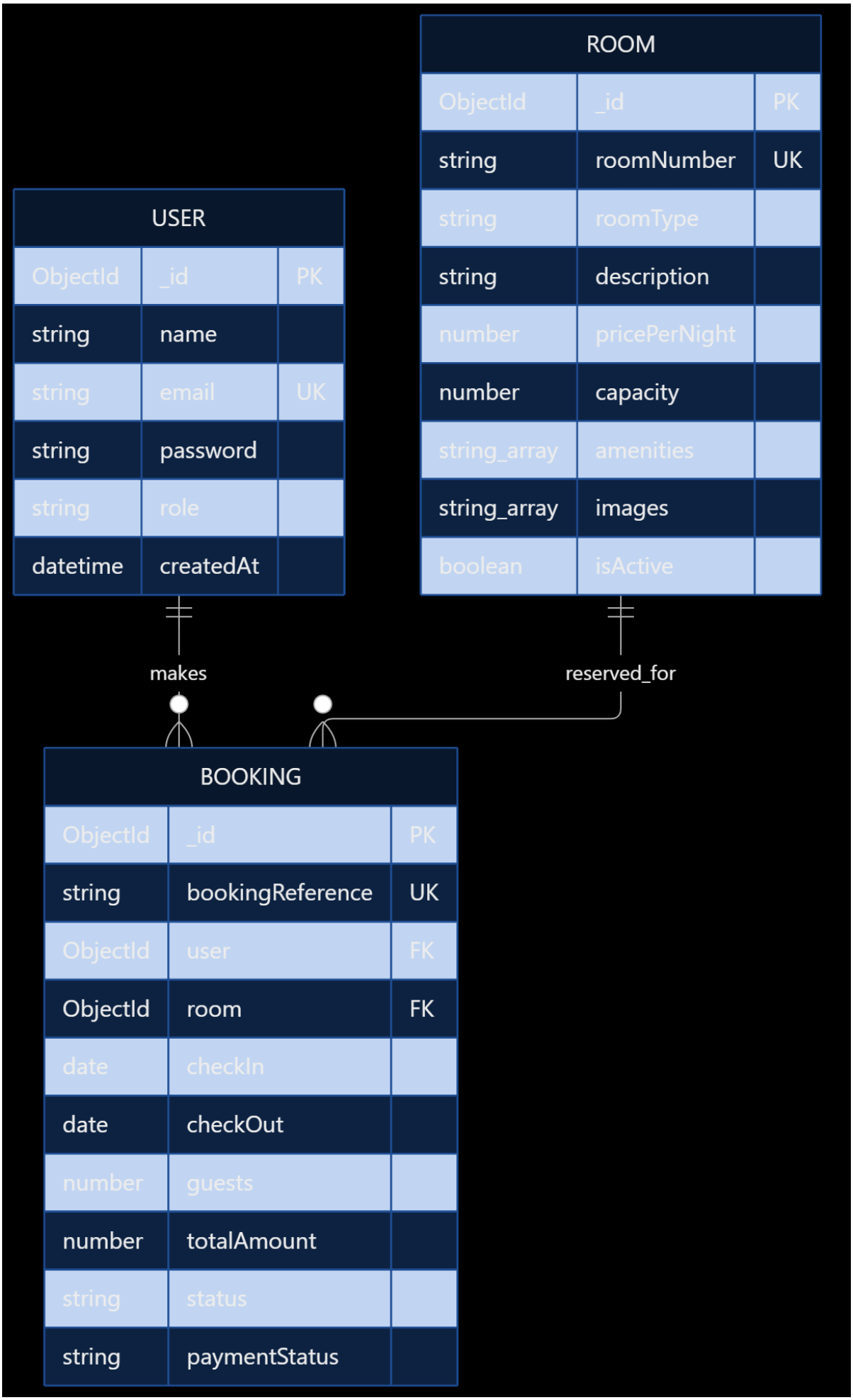
1. System Architecture

HotelPro three-tier architecture and authentication flow



2. Entity-Relationship (ER) Diagram

User, Room, and Booking entities with primary/foreign-key fields



3. Test-Case Report

Results below are based on tests reported during development. 'Passed (reported)' means the result was previously reported by the user or project conversation; these tests were not rerun during PDF creation.

ID	Test	Action / input	Expected / observed result	Status
TC-01	Registration	Register new customer	Customer account created; role forced to customer	Passed (reported)
TC-02	Customer login	Valid customer credentials	Authenticated; customer dashboard opens	Passed (reported)
TC-03	Admin login	Valid admin credentials	Admin redirected to /admin	Passed (reported)
TC-04	Room browsing	Open room list/details	Active rooms and details shown	Passed (reported)
TC-05	Booking creation	Valid dates and guest count	Booking created and total calculated	Passed (reported)
TC-06	Date conflict	Overlap a confirmed booking	Conflicting booking rejected	Passed (reported)
TC-07	Adjacent dates	Book when previous stay ends	Non-overlapping dates allowed	Passed (reported)
TC-08	Cancellation	Cancel own booking	Cancellation succeeds	Passed (reported)
TC-09	Room management	Admin add/edit/deactivate room	Admin changes work; inactive room hidden	Passed (reported)
TC-10	Inactive room	Try to book inactive room	Backend rejects request	Passed (reported)
TC-11	Admin pages	Open bookings and users pages	Pages load and show records	Passed (reported)
TC-12	Role restriction	Customer POST /api/rooms	Request blocked	Passed (reported)
TC-13	Admin-only endpoint	Customer GET /api/bookings/admin/all	Request blocked	Passed (reported)
TC-14	Invalid token	GET /api/profile with invalid JWT	401 Unauthorized	Passed (reported)
TC-15	Role escalation	Register with role=admin	Server assigns customer role	Passed (reported)
TC-16	Git secret check	Verify .env ignore/history	Must be checked locally	Pending

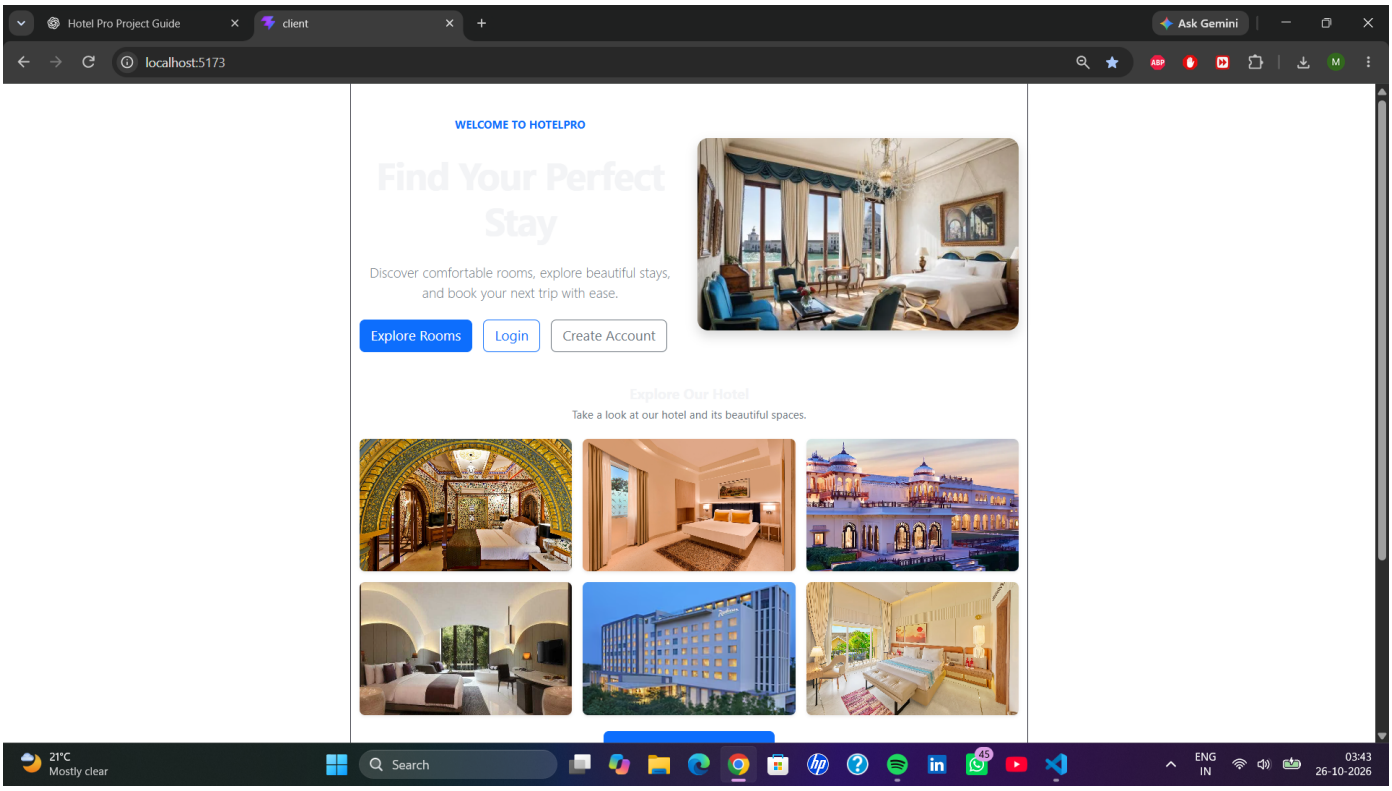
Security and limitations

The project uses JWT protected routes and admin role checks; passwords are expected to be hashed using bcrypt. Payment status is tracked, but actual payment-gateway processing/refunds are not confirmed as implemented. For production, enforce booking availability atomically to reduce simultaneous booking race conditions.

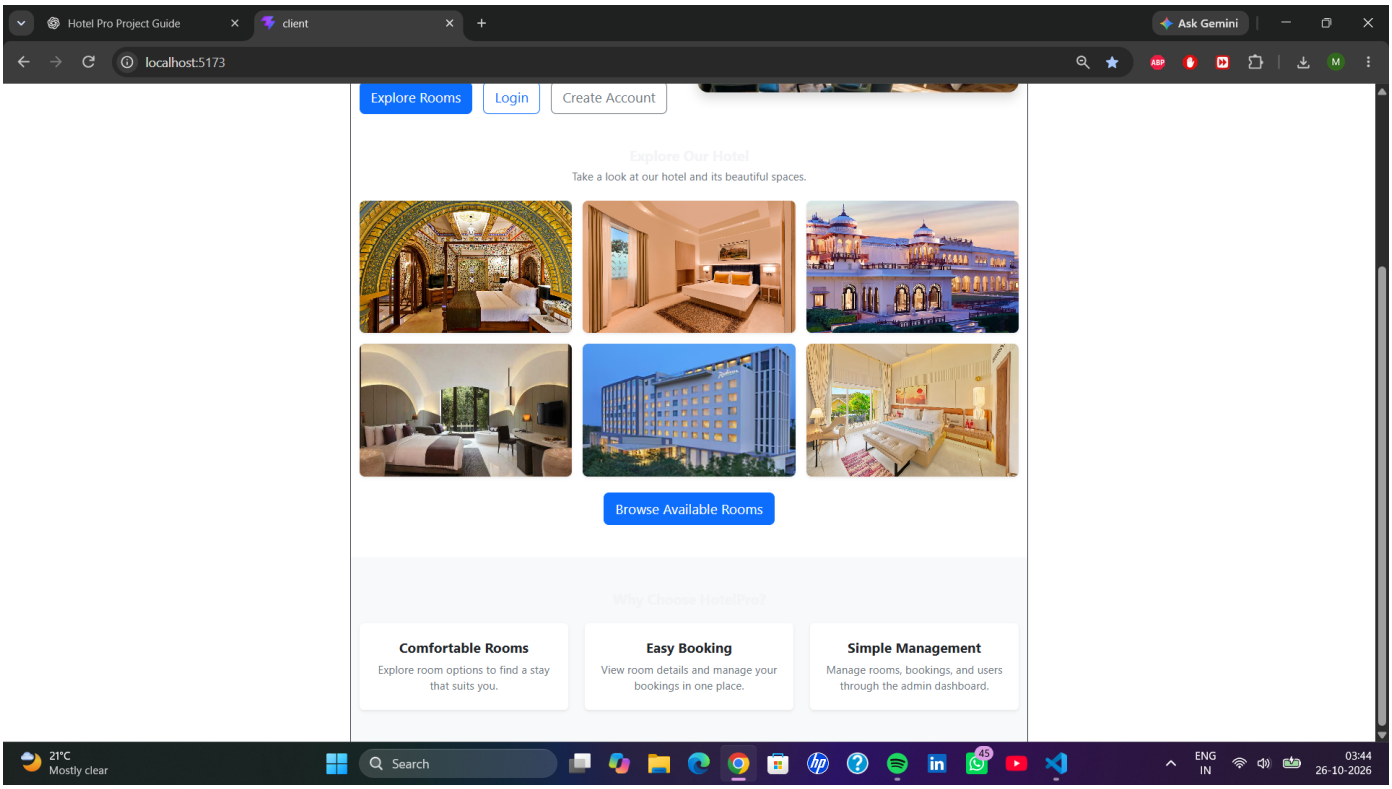
4. Application Screenshots

Two screenshots per page to use the page space more effectively

Screenshot 1: Homepage with hotel banner and gallery



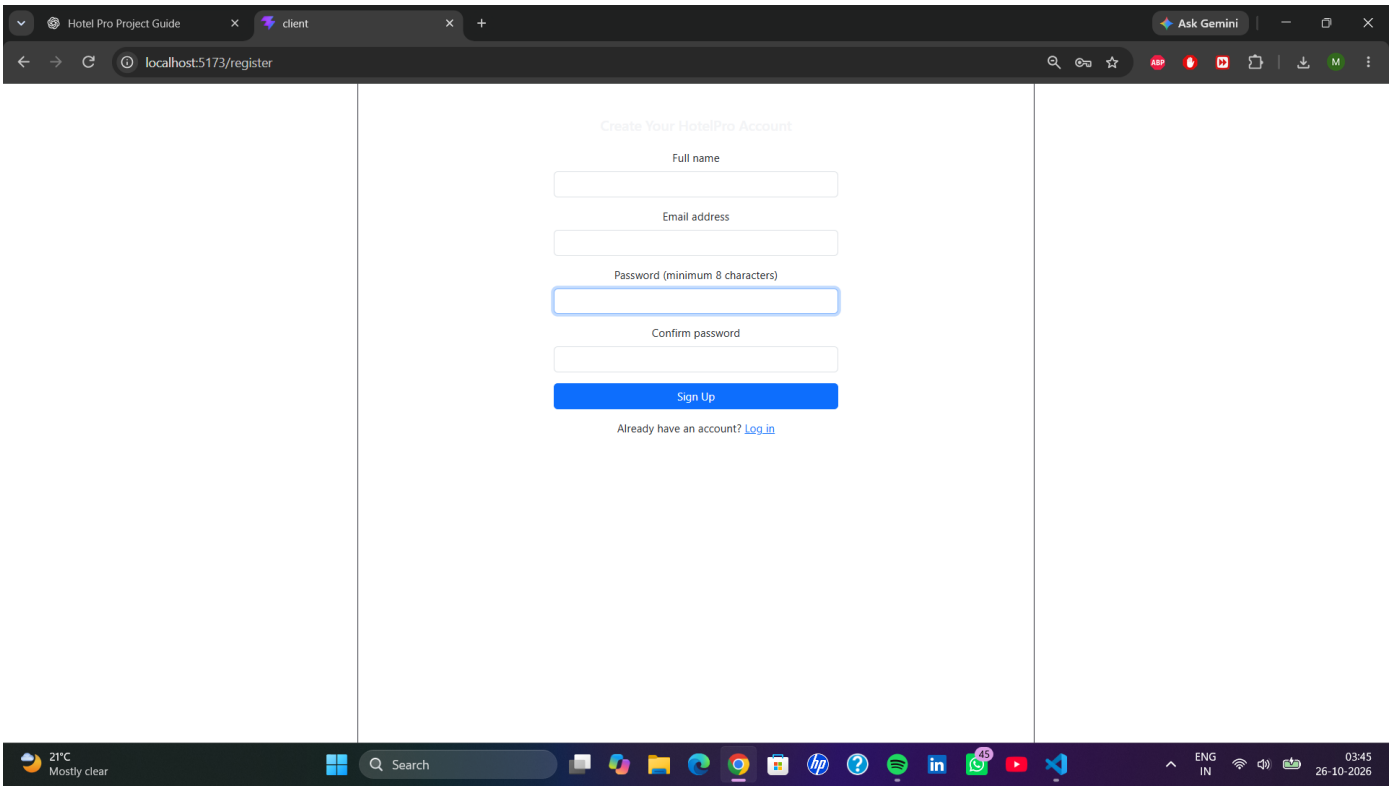
Screenshot 2: Homepage gallery and feature section



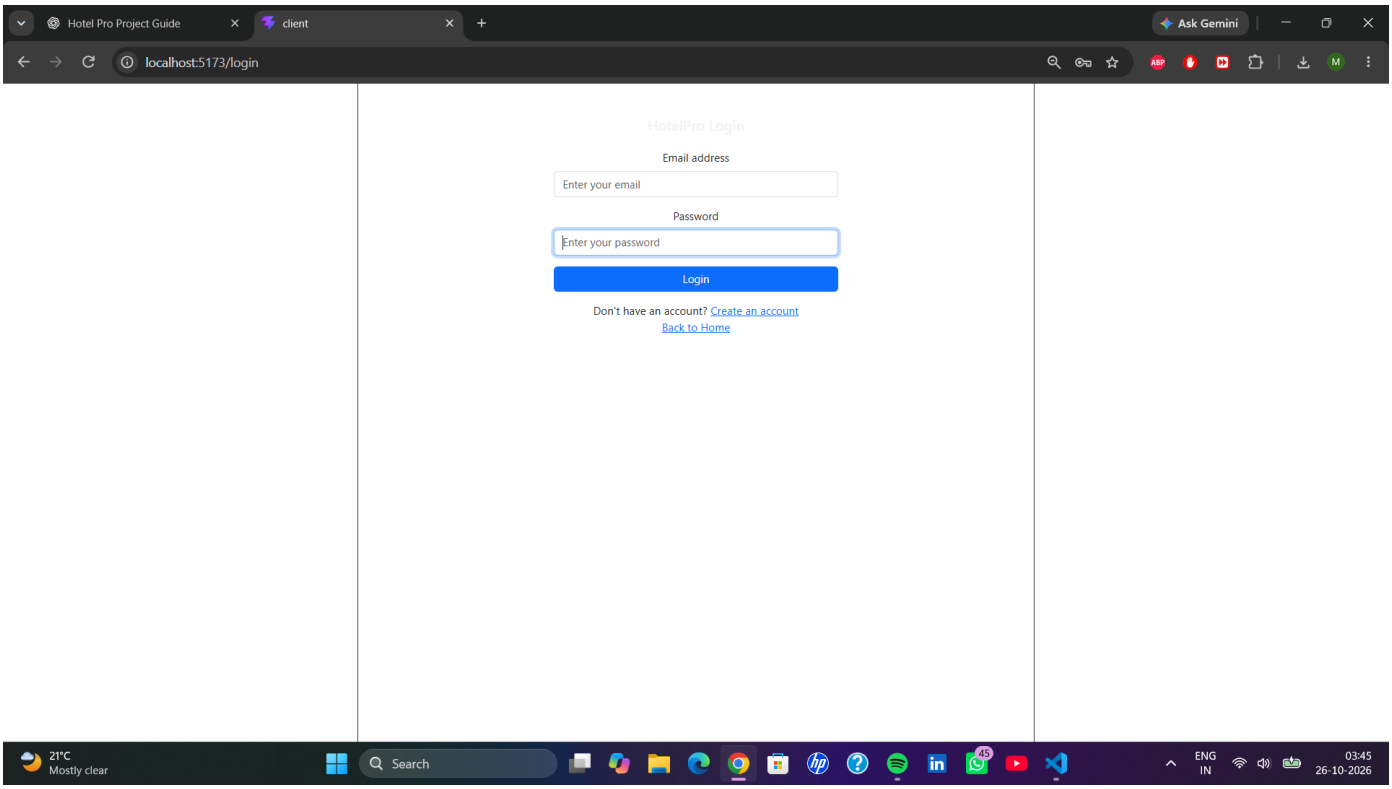
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Screenshot 3: Customer registration page



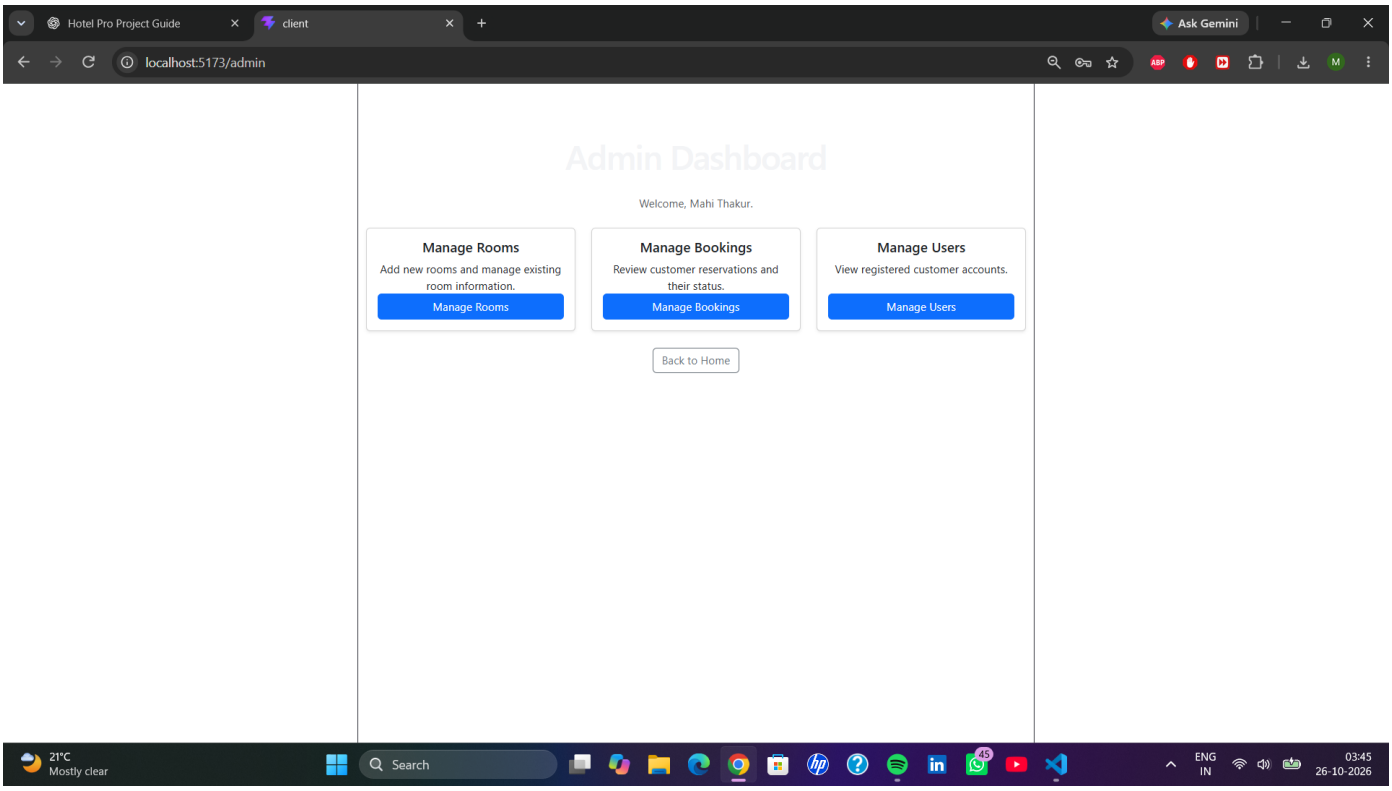
Screenshot 4: Customer login page



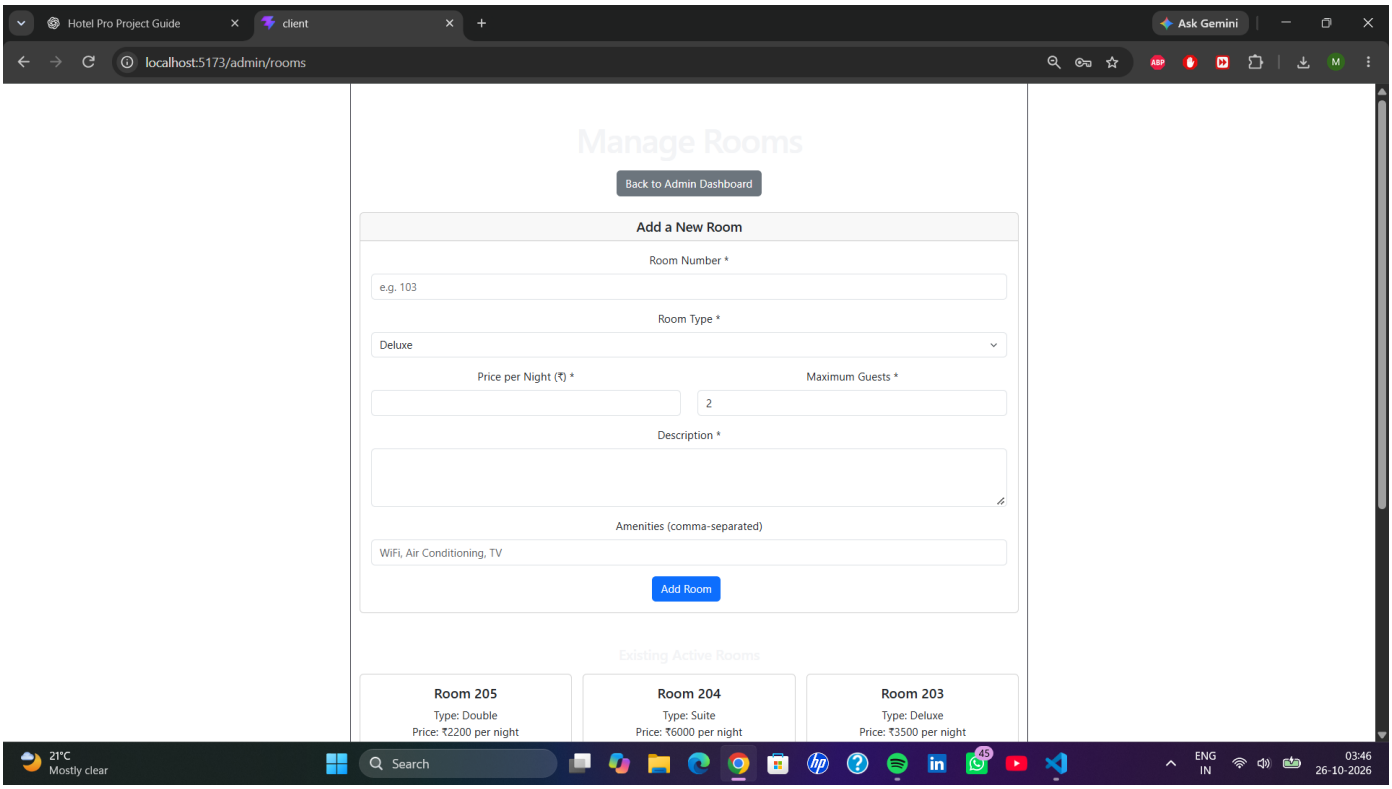
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Screenshot 5: Admin dashboard



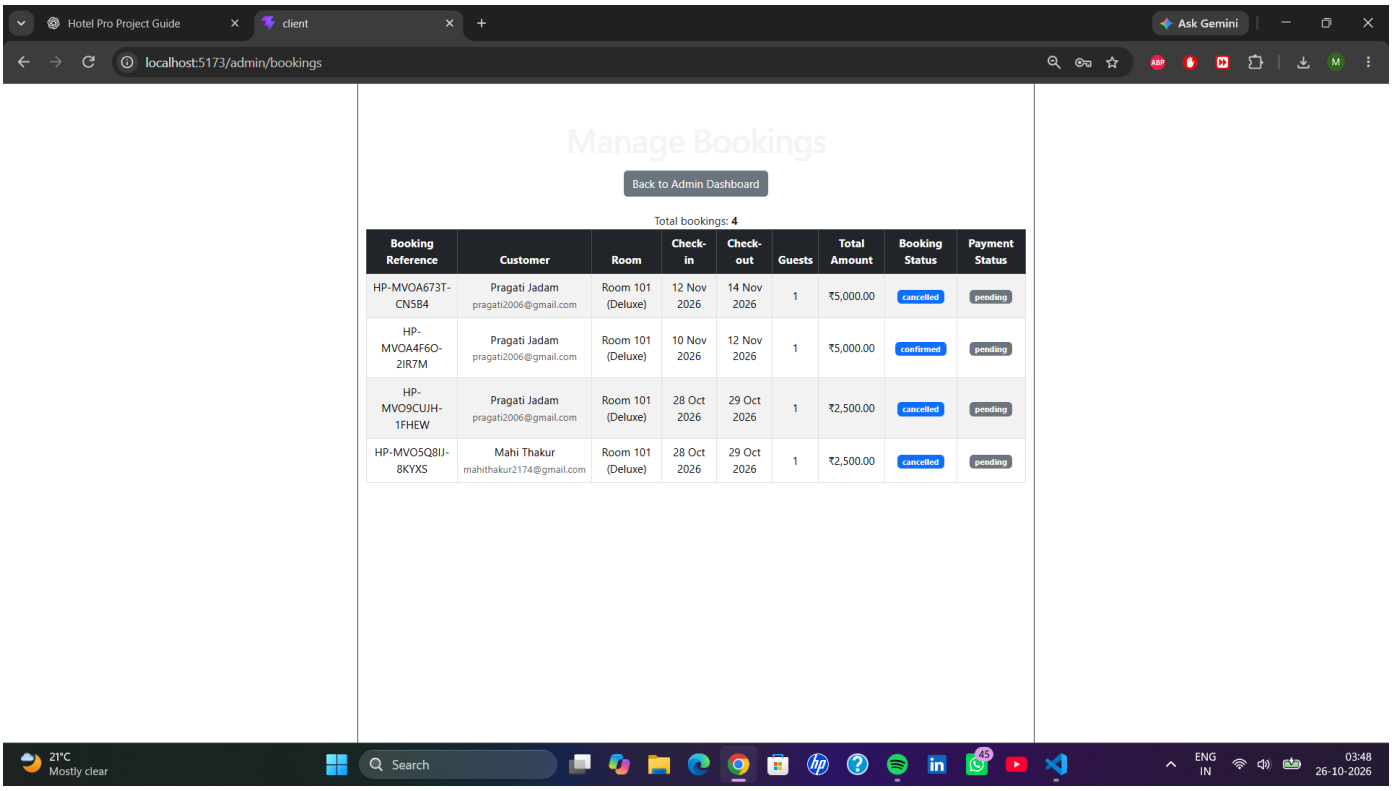
Screenshot 6: Admin Manage Rooms page



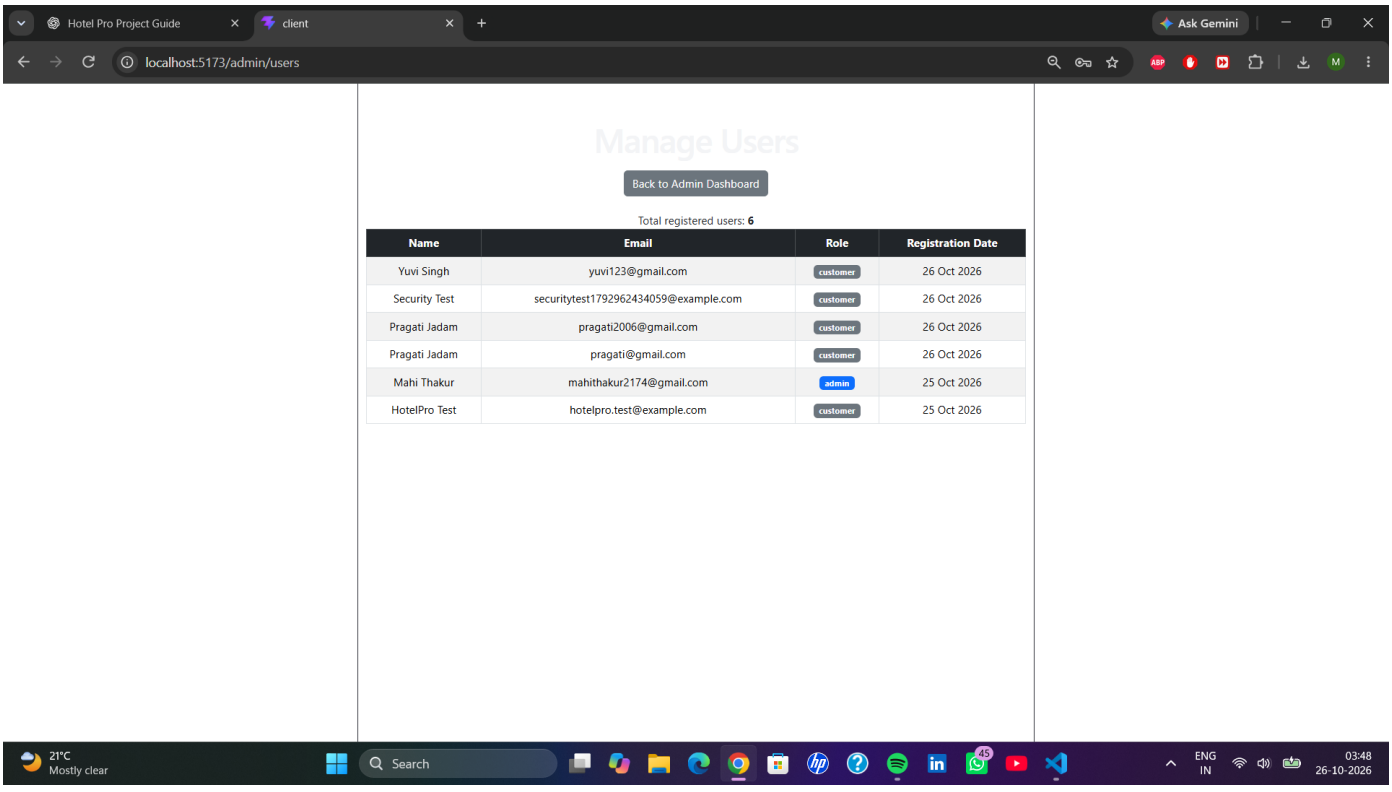
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Screenshot 7: Admin Manage Bookings page



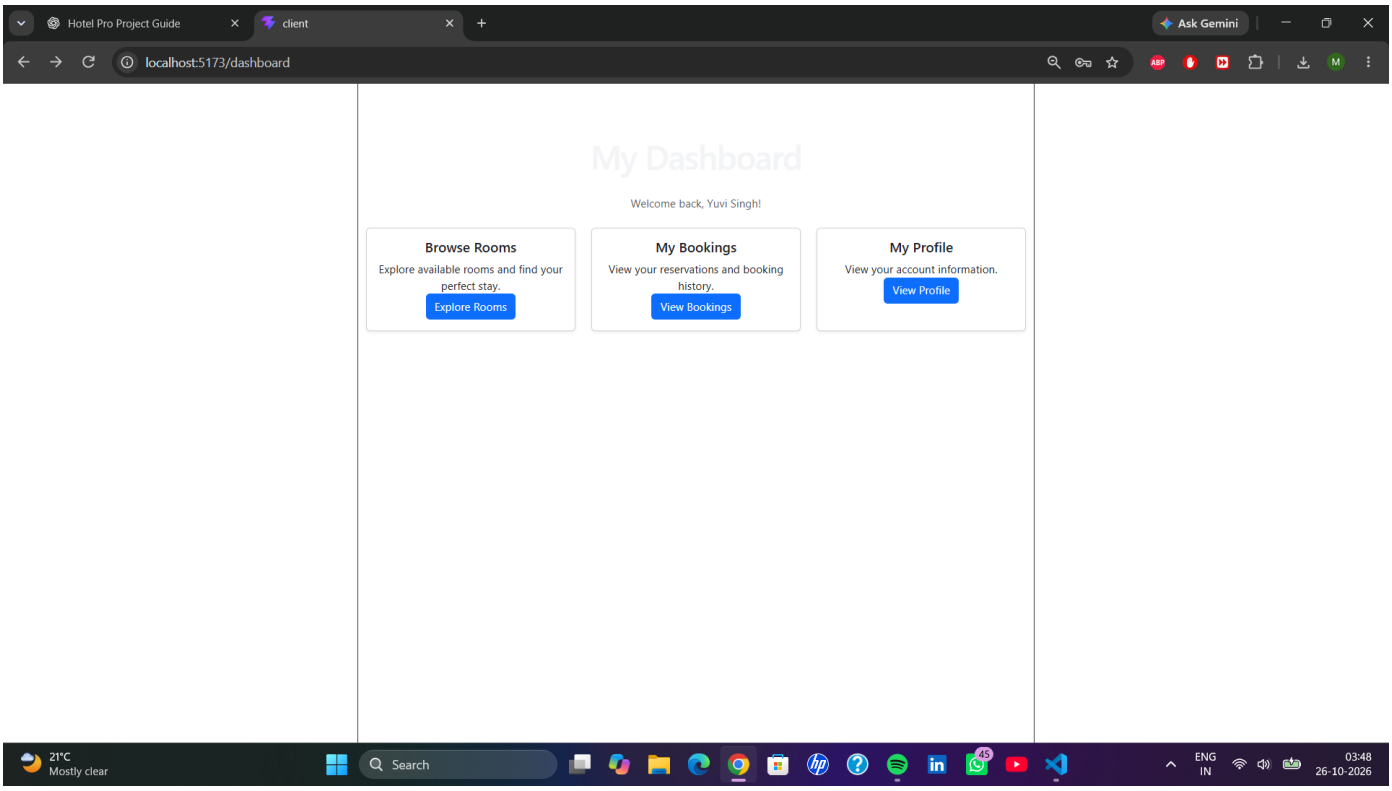
Screenshot 8: Admin Manage Users page



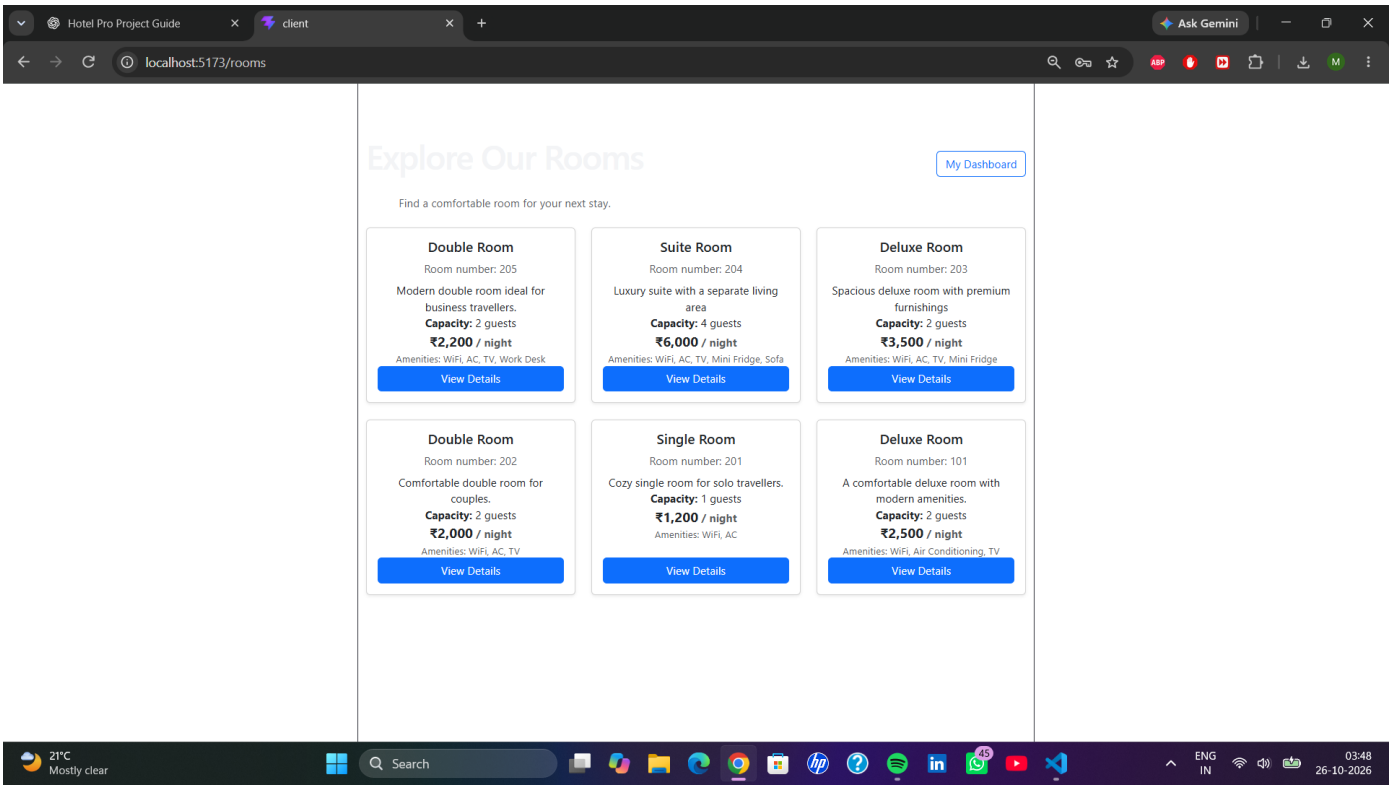
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Screenshot 9: Customer dashboard



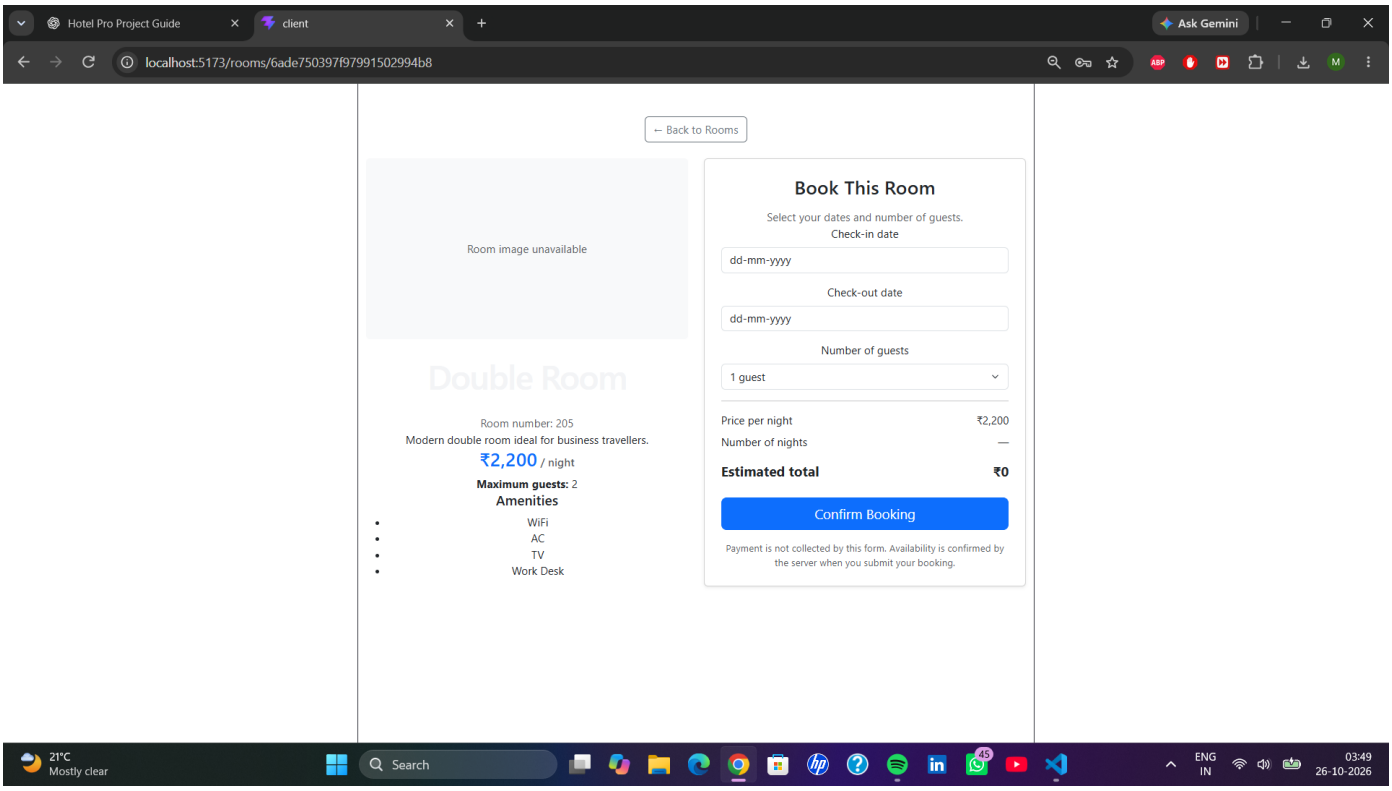
Screenshot 10: Room listing page



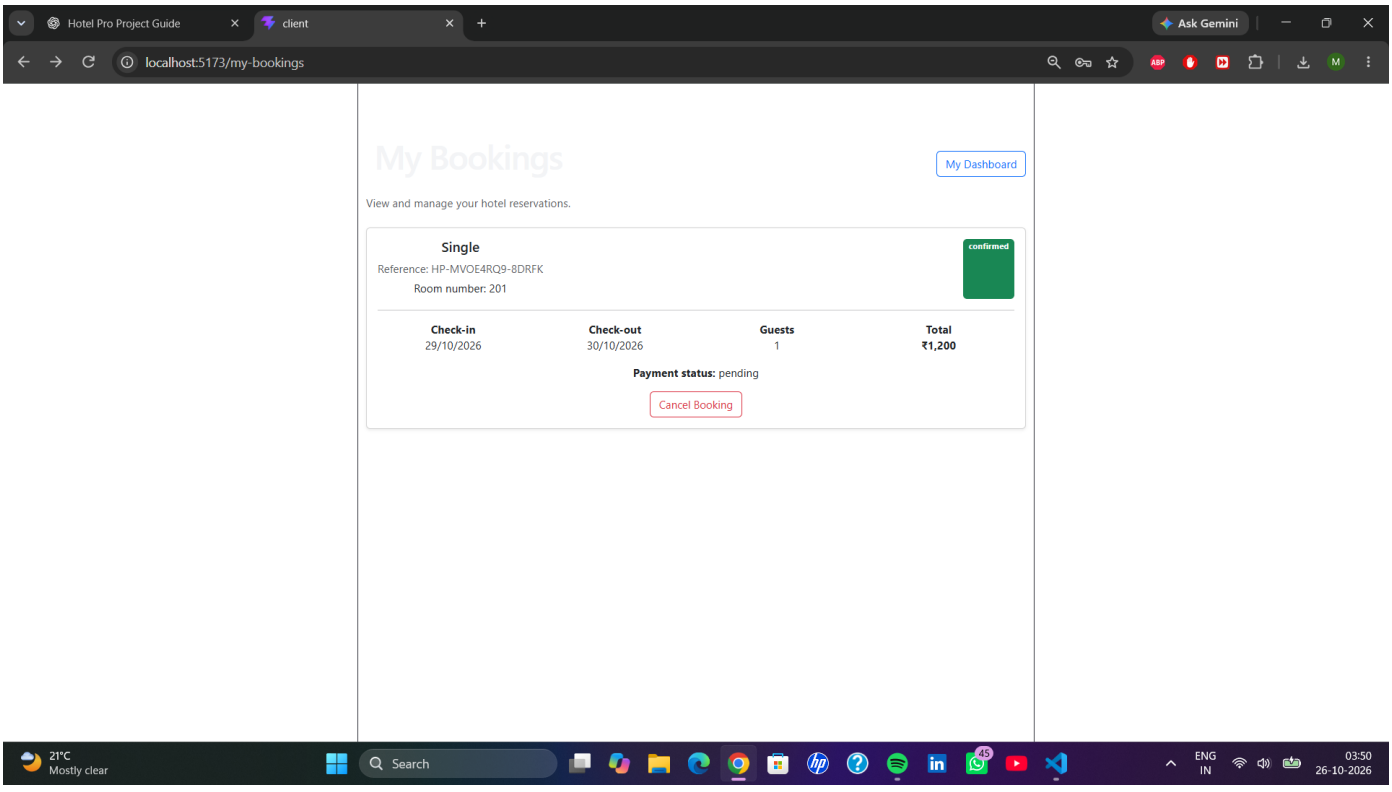
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Screenshot 11: Room details and booking form



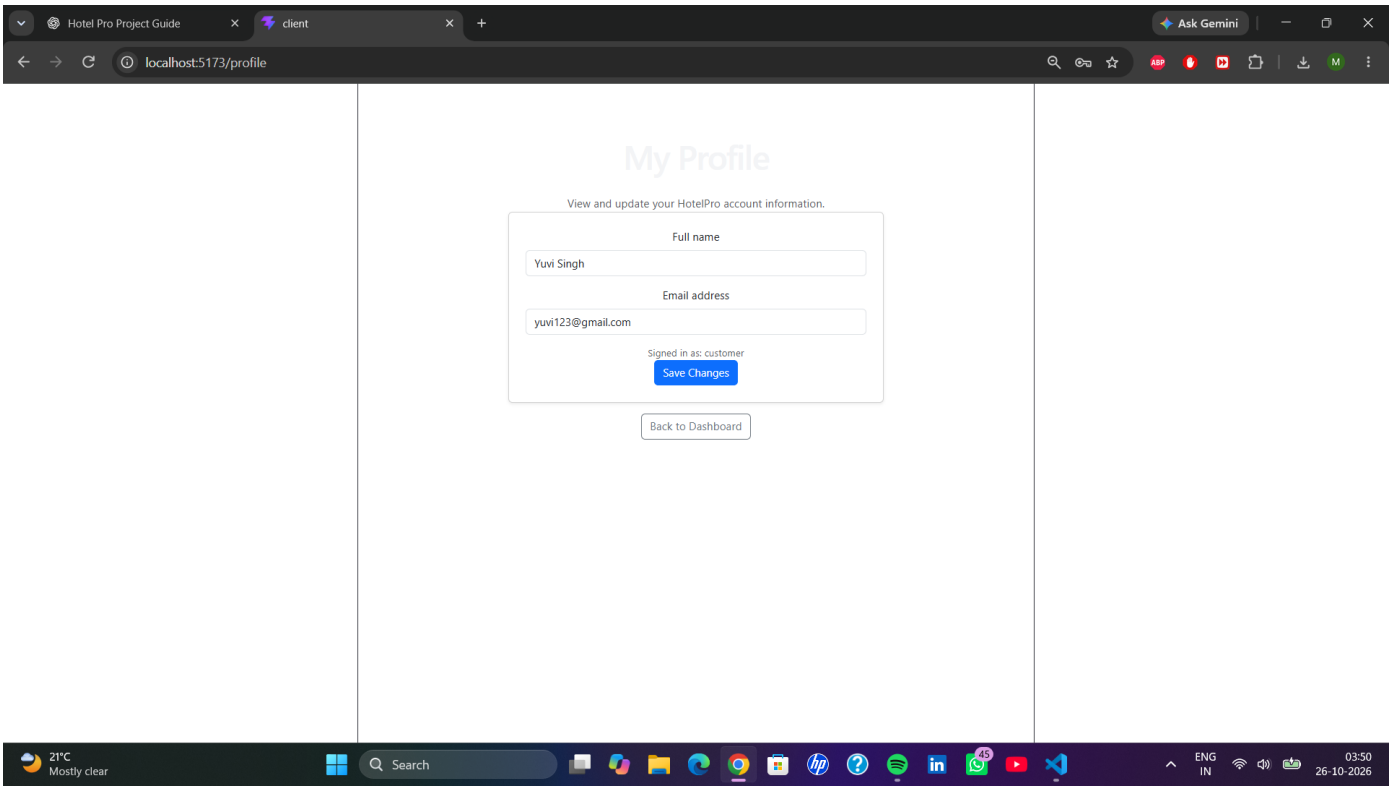
Screenshot 12: My Bookings page



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Screenshot 13: Customer profile page



5. Submission Checklist

Item	Status
ER diagram added to report	Included in this PDF
System architecture diagram added	Included in this PDF
Customer page screenshots	Included
All admin page screenshots	Included: dashboard, rooms, bookings, users
Test cases and results	Included; results labelled by evidence status
README installation/run commands	Not verified here
.env excluded from Git; no secrets committed	Pending local verification
Final demo video	Still to record / not supplied

Before submission: verify README commands, ensure .env is in .gitignore, inspect Git history for secrets, start client and server from clean terminals, and record a short customer booking/admin management demo.